Yong Guk Kang

Research Professor, Imaging Intelligence Laboratory, School of Mechanical Engineering, Seoul National University

Optical systems and computational imaging researcher at Seoul National University's Imaging Intelligence Laboratory, experienced in imaging apparatus design, physics-based simulation, rapid prototyping, and advanced image processing. Current focus includes wave-optics simulation, inverse design of optical elements such as phase masks and diffractive optics, lensless camera reconstruction, and machine-learning assisted solutions for ill-posed imaging problems.

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Skills

Computational optics: Wave optics simulation, Lensless imaging, Inverse design, Optical image reconstruction, Physics-based noise modeling, Interferometric imaging, Image deconvolution, Image correlation based strain mapping

Optical hardware: Multiphoton microscopy, Optical coherence tomography, Optical coherence microscopy, Off-axis holographic microscopy, Optical bench prototyping, Lensless cameras

Programming: MATLAB, Python, PyTorch, C++

Fabrication and prototyping: DMD grayscale lithography, PDMS microfluidic chip fabrication, FDM 3D printing, Fusion 360, ZEMAX, ImageJ

Experience

Research Professor, Seoul National University, School of Mechanical Engineering (2024-04-present)

- Designed and simulated transparent interactive display systems with integrated camera functionality.
- Developed lensless imaging systems and wave-optics simulation workflows.
- Applied machine learning based inverse design methods for optical elements.
- Fabricated optical elements using DMD light-engine based grayscale lithography workflows.

Postdoctoral Researcher, Korea University, School of Biomedical Engineering (2022-09-2024-03)

- Implemented successive accumulation of interferogram based 3D reflection phase microscopy.
- Developed sparse axial scanning and virtual refocusing strategies for efficient 3D imaging.
- Designed relay optics for laser scanning systems.
- Built ZEMAX-based 4-f relay optics from off-the-shelf components.

Research Lecturer, Korea University, School of Medicine (2020-10-2022-08)

- Developed optical coherence microscopy methods for quantifying cell-ECM interactions.

- Applied temporal dynamic contrast imaging to analyze intracellular motility.
- Simulated Bessel/Gaussian beam switchable OCM to extend depth of focus.

Projects

Lensless and HDR computational imaging. Development of lensless imaging systems for high dynamic range applications in extreme lighting conditions, including non-paraxial wave simulation and inverse design of optical elements.

Reflection phase microscopy. 3D reflective phase microscopy using successive accumulation of interferograms, sparse axial scanning, virtual refocusing, and field synthesis.

Optical coherence microscopy for cell-ECM dynamics. Optical coherence microscopy methods for live-cell and tissue imaging with temporal dynamic contrast.

Live-cell nonlinear microscopy for matrix deformation analysis. Second-harmonic generation and two-photon microscopy workflows for visualizing collagen matrix deformation and cell-ECM mechanical interactions.

Optical prototyping and fabrication. Rapid prototyping for optical and biomedical experiments, including FDM 3D printing, PDMS microfluidic chip fabrication, and DMD grayscale lithography for planar optical elements.

Education

Ph.D. in Biomedical Engineering, Korea University

B.S. in Biomedical Engineering, Korea University

Selected Publications

Three-dimensional imaging in reflection phase microscopy with minimal axial scanning, Optics Express (2023)

Optical coherence microscopy with a split-spectrum image reconstruction method for temporal-dynamics contrast-based imaging of intracellular motility, Biomedical Optics Express (2023)

Wide-area topography using reflection phase microscopy for surface inspection, Optics Express (2025)

Non-Paraxial Layered Diffraction Simulator for Inverse Design of Lensless Imaging Optics, Optica Imaging Congress 2025 (2025)

High-speed virtual re-staining via near-infrared quantitative phase imaging, Proceedings of SPIE Advanced Biophotonics Conference (2026)

Semi-supervised virtual staining using learned-illumination Fourier ptychography for high-speed label-free histopathology, Journal of Physics: Photonics (2025)

Cardiac fibroblast-mediated ECM remodeling regulates maturation in an in vitro 3D engineered cardiac tissue, Journal of Tissue Engineering (2025)

Development of a 3-D Physical Dynamics Monitoring System Using OCM with DVC for Quantification of Sprouting Endothelial Cells Interacting with a Collagen Matrix, Materials (2020)

Quantification of focal adhesion dynamics of cell movement based on cell-induced collagen matrix deformation using second-harmonic generation microscopy, Journal of Biomedical

Optics (2018)

Laser-tissue interaction simulation considering skin-specific data to predict photothermal damage lesions during laser irradiation, Journal of Computational Design and Engineering (2023)

Reflection Phase Microscopy by Successive Accumulation of Interferograms, ACS Photonics (2019)

Single-shot digital holographic microscopy for quantifying a spatially-resolved Jones matrix of biological specimens, Optics Express (2016)

Patents and Recognition

Funding: Creative and Challenging Research Base Support (2021R1I1A1A01048370)

Fellowship: Global Ph.D. Fellowship (2015H1A2A1030048)

Patent: Head mount system for providing surgery support image (US 11,356,651)

Patent: Reflection phase microscopy system for 3D image acquisition (10-2024-0081557)

Patent: Phase mask fabrication method based on grayscale lithography (10-2024-0161144)